

## EXECUTIVE SUMMARY

This analysis was conducted to identify LGAs that may be at a higher risk of election irregularities in future elections. To achieve this, polling unit election data was merged with election incident reports from the 2019 and 2023 election cycles. The datasets were cleaned, standardized, and matched using State, LGA, and Election Year. During the cleaning process, a negative vote value (-10) was identified. Since vote counts cannot be negative in a real election, the affected record was treated as a data quality issue and excluded from the analysis. As a result, the final analysis was conducted using 999 valid polling unit records.

Additional metrics such as Turnout Percentage, Anomaly Flags, Incident Count, Average Severity Score, and Risk Score were calculated. A Risk Score model was developed using the formula:

$$\text{Risk Score} = (\text{Anomaly Rate} \times 0.4) + (\text{Normalized Severity} \times 0.3) + (\text{Normalized Incident Count} \times 0.2) + (\text{Turnout Anomaly} \times 10)$$

This formula was chosen because anomaly rates were considered the strongest indicator of potential election irregularities and were therefore assigned the highest weight. Incident severity and incident count were included to capture both the seriousness and frequency of reported incidents, while turnout anomalies helped identify unusual voting patterns.

The analysis identified 210 anomalous polling units out of 999 valid polling units, representing an anomaly rate of approximately 21%. A total of 200 election-related incidents were recorded across the study period. Risk Scores ranged from 29.77 to 63.19, indicating the presence of low-, medium-, and high-risk LGAs. The highest-risk LGAs recorded relatively high anomaly rates, incident counts, and severity scores, making them priority areas for election monitoring and intervention.

A comparison of the 2019 and 2023 election cycles revealed that Risk Scores increased across all analyzed LGAs. This suggests a deterioration in election risk indicators over time, with 2023 showing higher levels of risk than 2019. No LGA recorded a reduction in Risk Score during the period under review. In addition, all LGAs in the dataset reported incidents, meaning there were no cases of high anomalies combined with zero incident reports.

Based on these findings, INEC and election observer organizations should prioritize monitoring efforts in the highest-risk LGAs like Nasarawa by increasing observer coverage, strengthening voter education, and improving incident reporting systems. One limitation of this analysis is that the Risk Score is based on a simplified weighting model and may not capture every factor affecting election integrity. Nevertheless, the analysis provides a practical and data-driven approach for identifying potential election hotspots and supporting more targeted election monitoring efforts.